



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CLOUD SQL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

Q1 What is Cloud SQL and what is its primary purpose in its cloud ecosystem?

Q6 How do you manage configuration and secrets for Cloud SQL?

Q2 How does IAM work with Cloud SQL?

Architecture

Q7 What are the main cost drivers for Cloud SQL and how do you optimize them?

Q3 How do you monitor Cloud SQL in production?

Configuration

Q8 How do you implement high availability with Cloud SQL?

Compliance

Q4 How do you scale Cloud SQL to handle variable load?

Hardening

Q9 How does Cloud SQL integrate with a CI/CD pipeline?

Testing

Q5 How do you secure Cloud SQL in production?

SSO / IdP

Q10 What are common pitfalls when first adopting Cloud SQL?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Cloud SQL is a managed cloud service

Mitigation

Cloud SQL is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A6 Store sensitive values in a managed secret

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A2 Access to Cloud SQL is controlled via

Primitives

Access to Cloud SQL is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A7 Cost in Cloud SQL typically comes from

Automation

Cost in Cloud SQL typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A3 Cloud SQL emits metrics to the cloud

Hardening

Cloud SQL emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A8 Deploy Cloud SQL across multiple availability zones.

Governance

Deploy Cloud SQL across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A4 Cloud SQL supports auto-scaling policies based on

Gotchas

Cloud SQL supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A9 CI/CD pipelines use the cloud CLI or

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to Cloud SQL after tests pass. Infrastructure-as-code (Terraform, CDK) defines Cloud SQL configuration as version-controlled code deployed via the pipeline.

A5 Place Cloud SQL in a private subnet

Federation

Place Cloud SQL in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A10 Common mistakes include misconfigured IAM policies (too

Automation

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.